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(globals)

[*] CODE 9.cpp

```
1  /*Q9: Write a program to calculate simple and compound interest for given principal, rate, and time.
2
3  /*
4  Sample Test Cases:
5  Input 1:
6  1000 5 2
7  Output 1:
8  Simple Interest=100, Compound Interest=102.5
9
10 Input 2:
11 5000 7 3
12 Output 2:
13 Simple Interest=1050, Compound Interest=1125.76
14
15 */
16 #include <stdio.h>
17 #include <math.h>
18
19 int main() {
20     double principal, rate, time, si, ci;
21     printf("Enter principal amount: ");
22     scanf("%lf", &principal);
23
24     printf("Enter annual interest rate (in percentage): ");
25     scanf("%lf", &rate);
26
27     printf("Enter time period (in years): ");
28     scanf("%lf", &time);
29
30     // Calculate Simple Interest: (P * R * T) / 100
31     si = (principal * rate * time) / 100;
32
33     // Calculate Compound Interest: P * (1 + R / 100)^T - P
34     ci = principal * pow((1 + rate / 100), time) - principal;
35     printf("\nSimple Interest = %.2lf\n", si);
36     printf("Compound Interest = %.2lf\n", ci);
37
38     return 0;
39 }
```

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```
Enter principal amount: 12000
Enter annual interest rate (in percentage): 4
Enter time period (in years): 1
```

```
Simple Interest = 480.00
Compound Interest = 480.00
```

```
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Process exited after 20.52 seconds with return value 0
Press any key to continue . . . |
```